

Statement of the Maldacena Conjecture:

AdS: 10D Type IIB on $AdS_5 \times S^5$

$\rightarrow$ 5-form flux through S^5 and is an integer N equal to the radius of L of AdS_5

$$L^4 = 4\pi g_s N \alpha'$$

SYM: $D=4$; Maximal SUSY $N=4$, $G = SU(N)$
 $g_{YM}^2 = g_s$

• t' Hooft limit of SYM

$\lambda \equiv g_{YM}^2 N$ is fixed

for $N \rightarrow \infty$

$\cong$ Classical S.T. on $AdS_5 \times S^5$

Make note of this

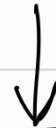
$\lambda \rightarrow \infty$: Reduces classical S.T. to Type IIB to classical sugra

$$\boxed{\begin{array}{l} \text{SYM} \\ \lambda = g_{YM}^2 N \end{array}}$$

$N \rightarrow \infty$
 $\xrightarrow{\text{weak coupling}}$

$$\boxed{\begin{array}{l} \text{Classical S.T. on} \\ AdS_5 \times S^5 \end{array}}$$

$\lambda \rightarrow \infty$



Classical Type IIB Sugra

* Super symmetry & Gauge theories: $D=4$

$$M_{\mu\nu} = (-, +, +, +), \mu, \nu = 0, 1, 2, 3$$

* Poincare' $\equiv \mathbb{R}^4 \ltimes SO(1, 3)$

Lorentz: $SO(1, 3)$; $L_{\mu\nu}$

Translation: $\mathbb{R}^4$; P_μ

$$SU(1, 3)^\mathbb{C} \cong SU(2) \times SU(2)$$

Spin $1/2 \uparrow$ Spin $1/2 \downarrow$

Reps of $SU(1, 3)^\mathbb{C} \cong SU(2) \times SU(2) \ni (S, S)$

labelled by their dim rep of these groups.

- Vector rep: $\underline{4} \in SO(1,3) \cong (\frac{1}{2}, \frac{1}{2})$
 $s_+, s_- \in \mathbb{Z}/2$ $\sum (s_+, s_-) = \text{Spin in } SO(1,3)$
- Scalar: $\underline{1}$ of $SO(1,3) \cong (0,0)$
- Symm. rank 2-tensor:

$$4 \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} \quad \frac{n^2 - n}{2} + \frac{2n}{2} = \frac{n^2 + n}{2}$$

Symm rank 2-tensor has $n(n+1)/2$

D.O.F. $n=4 \Rightarrow 10$

$$\underline{10} \in \Delta(1,3)^{\oplus} \cong (1,1)$$

○—○

- (i) Left chirality: $(\frac{1}{2}, 0)$
- (ii) Right chirality: $(0, \frac{1}{2})$
- (iii) Self dual Sym^2 : $(1, 0)$
- (iv) Anti-S.D. Sym^2 : $(0, 1)$

$a = 1, \dots, N$

$$\begin{cases} Q_\alpha^a & \alpha=1,2 \text{ left weyl} \\ \bar{Q}_{\dot{\alpha}a} = (Q_\alpha^a)^\dagger & \text{right weyl} \end{cases}$$

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \quad Q^a = \begin{pmatrix} Q_\alpha^a \\ \bar{Q}_{\dot{\alpha}a} \end{pmatrix}$$

- Susy algebra

$$\{Q_\alpha^a, \bar{Q}_{\dot{\beta}b}\} = 2\sigma_{\alpha\dot{\beta}}^\mu P_\mu \delta_b^a$$

$$\{Q_\alpha^a, Q_\beta^b\} = 2\epsilon_{\alpha\beta} Z^{ab}$$

Z : Central charges

$$Z^{ab} = -Z^{ba}$$

$$[Z^{ab}, \text{Anything}] = 0$$

For $N=1, a=b=1$

$$\underline{Z^{11}} = -Z^{22} \Rightarrow Z = 0$$

* Massless Particle Reps: $P_\mu = (E, 0, 0, E), E > 0$

$$\{Q_\alpha^a, \bar{Q}_{\dot{\beta} b}\} = 2(\sigma_{\alpha\dot{\beta}}^\mu P_\mu) \delta_b^a$$

$$= 2(\sigma_{\alpha\dot{\beta}}^0 P_0 + \dots + \sigma_{\alpha\dot{\beta}}^3 P_3) \delta_b^a$$

$$= (-2E\sigma_{\alpha\dot{\beta}}^0 + 2E\sigma_{\alpha\dot{\beta}}^3) \delta_b^a$$

$\sigma^\mu = (-1, \vec{\sigma})$

$$= 2E\delta_b^a \left(-\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right) \alpha_{\dot{\beta}}$$

$$= \delta_b^a \begin{pmatrix} 4E & 0 \\ 0 & 0 \end{pmatrix} \alpha_{\dot{\beta}}$$

$$\{Q_\alpha^a, \bar{Q}_{\dot{\beta} b}\} = \delta_b^a \begin{pmatrix} 4E & 0 \\ 0 & 0 \end{pmatrix} \alpha_{\dot{\beta}}$$

* Upon Quant. we look at Q and $Q^\dagger$ as ops on a par. def. Hilbert Space of States:

(b) $\alpha = \dot{\beta} = 2$ and $a = b$

$$\{(Q_2^a), (Q_2^a)^\dagger\} = 0$$

$$\therefore \langle \psi | (Q_2^a Q_2^{\dagger a} + Q_2^{\dagger a} Q_2^a) | \psi \rangle = 0$$

$$\langle \psi | Q_2^a Q_2^{\dagger a} | \psi \rangle + \langle Q_2^a \psi | Q_2^a \psi \rangle = 0$$

$$\underbrace{\langle Q_2^a \psi | Q_2^a \psi \rangle}_{> 0} = 0$$

But $\langle 1 \rangle \neq 0$ so we must have

$$Q_2^a = 0$$

as such, $\{Q_\alpha^a, Q_\beta^b\} = 2\epsilon_{\alpha\beta} Z^{ab}$

$\alpha = 2, \beta = 1$

$$\{Q_2^a, Q_1^b\} = 2(-1) Z^{ab}$$

$$= -2 Z^{ab}$$

$$\therefore Z^{ab} = 0$$

* Q_1^a : lowers helicity by $1/2$
 $\bar{Q}_1^a = (Q_1^a)^\dagger$ Raises helicity by $1/2$

$[Q_\alpha^a, \bar{Q}_\beta^a] = \delta_{\alpha\beta} \delta^a_b$ Form a rep of dim 2^N of the Clifford alg. associated w the Lie alg. $SO(2N)$.

Massive Particle Reps: $P^\mu = (M, 0, 0, 0)$

$\{Q_\alpha^a, (\bar{Q}_\beta^b)^\dagger\} = 2M \delta_\alpha^\beta \delta^a_b$
 $\rightarrow$ Make use of $SU(N)_R$ symmetry to diagonalize in blocks of 2×2 anti-symm matrices $Z^{ab} = -Z^{ba}$

* Split $a \equiv (\hat{a}, \bar{a})$ $\hat{a} = 1, 2, \dots, r$
 $\bar{a} = 1, \dots, r$
 $N = 2r$ (N is even)

$\rightarrow Z = \text{diag}(eZ_1, \dots, eZ_r, \#)$

$$e^{12} = -e^{21} = 1 \Rightarrow e^{ab} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\# = \begin{cases} 0 & N = 2r+1 \\ \text{absent} & N = 2r \end{cases}$$

$Z_{\bar{a}, \hat{a}} = 1, \dots, r$ REAL Central charges

$$Z^{ab} = \begin{pmatrix} 0 & -a & \dots & -N-1 \\ a & & & \\ \vdots & & \ddots & \\ N-1 & \dots & & 0 \end{pmatrix}$$

$\exists A$ s.t.

$$Z \equiv A Z^{ab} A^{-1}$$

If we have $SU(N)_R \supset G$

$$G = e^{T\lambda} \text{ then } [Q, T] = 0$$

$\therefore$

$$G \{Q, Q\} G^{-1} G|4\rangle = G 2 \epsilon_{\alpha\beta} Z^{ab} |4\rangle$$

$$\therefore G \{Q_\alpha, Q_\beta\} G^{-1} = 2 \epsilon_{\alpha\beta} Z^{ab}$$

$$\epsilon^{\gamma\alpha} \frac{1}{2} G \{Q_\alpha^a, Q_\beta^b\} G^{-1} = \epsilon^{\gamma\alpha} \epsilon_{\alpha\beta} Z^{ab}$$

$$\epsilon^{\gamma\alpha} \frac{1}{2} G \{Q_\alpha^a, Q_\beta^b\} G^{-1} = \delta_B^{\gamma\alpha} Z^{ab}$$

Define: $Q_{\alpha\pm}^{\bar{a}} \equiv \frac{1}{2} (Q_\alpha^{\pm\bar{a}} \pm \sigma_{\alpha\beta}^0 (Q_\beta^{\mp\bar{a}})^\dagger)$

$$\{Q_{\alpha\pm}^{\bar{a}}, (Q_{\beta\pm}^b)^\dagger\} = \delta_{\alpha\beta}^{\bar{a}b} (M \pm Z_{\bar{a}})$$

* For this to be a unitary particle rep $\Delta H \{Q, Q\} |4\rangle > 0$

We obtain the "famous" **BPS**

Bound giving a lower bound on the mass in terms of the central charges

$$M \geq |Z_{\bar{a}}|, \quad \bar{a} = 1, \dots, r = \lfloor N/2 \rfloor$$

* If $|Z_{\bar{a}}| = M$ for some $\bar{a}$, then the BPS bound is saturated and either the supercharges $Q_{\alpha+}^{\bar{a}}$ or $Q_{\alpha-}^{\bar{a}}$ must vanish

$$\rightarrow M = |Z_{\bar{a}}| \text{ for } \bar{a} = 1, \dots, r_0$$

$$M > |Z_{\bar{a}}| \text{ for } \bar{a} = r_0 + 1, \dots, r$$

Then the susy alg. is effectively

a Clifford Alg. associated w/
 $SO(4N - 4r_0)$

The corresponding reps are said to be
 $1/2^{\text{rd}}$ BPS w dim 2^{2N-2r_0}

Aside : $\{Q_\alpha^a, \bar{Q}_\beta^b\} = 2\epsilon_{\alpha\beta} Z^{ab}$
 $a = (\hat{a}, \bar{a})$

$$\{Q_\alpha^{(\hat{a}, \bar{a})}, \bar{Q}_\beta^{(\hat{b}, \bar{b})}\} = 2\epsilon_{\alpha\beta} Z^{(\hat{a}, \bar{a})(\hat{b}, \bar{b})}$$

$$\{Q_\alpha^{\bar{a}\hat{a}}, \bar{Q}_\beta^{\bar{b}\hat{b}}\} = 2\epsilon_{\alpha\beta} Z^{(\hat{a}, \bar{a})(\hat{b}, \bar{b})}$$

Define, $Q_{\alpha\pm}^{\bar{a}} = \frac{1}{2}(Q_\alpha^{\pm\bar{a}} \pm \sigma_{\alpha\beta}^0 (Q_\beta^{2\bar{a}})^\dagger)$

$$\{Q_\alpha^a, Q_\beta^b\} = 2\sigma_{\alpha\beta}^\mu P_\mu \delta_b^a$$

Then

$$\begin{aligned} \{Q_{\alpha\pm}^{\bar{a}}, (Q_{\beta\pm}^{\bar{b}})^\dagger\} &= \frac{1}{4} \{Q_\alpha^{\pm\bar{a}} \pm \sigma_{\alpha\beta}^0 (Q_\beta^{2\bar{a}})^\dagger, \\ &\quad Q_\beta^{\pm\bar{b}} \pm \sigma_{\alpha\beta}^0 (\bar{Q}^{2\bar{b}})^\dagger\} \\ &= \frac{1}{4} \{Q_\alpha^{\pm\bar{a}} Q_\beta^{\pm\bar{b}}\} \pm \{ \sigma_{\alpha\beta}^0 \bar{Q}^{2\bar{a}} \bar{Q}_\beta^{\pm\bar{b}}, Q_\beta^{\pm\bar{b}} \} \\ &\quad \pm \{Q_\alpha^{\pm\bar{a}}, \sigma_{\alpha\beta}^0 (\bar{Q}^{2\bar{b}})^\dagger\} \pm \sigma_{\alpha\beta}^0 \sigma_{\alpha\gamma}^0 \sigma_{\beta\delta}^0 Q_\gamma^{\pm\bar{b}} \dots \\ &= \frac{1}{4} (2\epsilon_{\alpha\beta} Z^{(\bar{a})(\bar{b})}) \end{aligned}$$

$$\Rightarrow Z^{ab} = (Z_{\bar{a}})^{\hat{a}\bar{b}}$$

$$\therefore Z_{\bar{a}} \equiv (Z_{\bar{a}})^{\hat{a}\bar{b}}$$

$$(Z_{\bar{a}}) = \begin{pmatrix} Z_{\bar{a}}^{11} & & Z_{\bar{a}}^{1N} \\ Z_{\bar{a}}^{21} & \dots & Z_{\bar{a}}^{2N} \\ \vdots & & \vdots \\ Z_{\bar{a}}^{N1} & & Z_{\bar{a}}^{NN} \end{pmatrix}$$

* Anti-symm: $Z^{ab} = -Z^{ba}$

$$(1^2 - n)/2 = \frac{n(n-1)}{2} \text{ elements}$$

$$Z_{\bar{a}} : (2N - 2) = 2(N - 1)$$

$$\sum_{x=1}^N 2(x-1) = 2 \sum_{x=1}^N x - 2N$$

$$= \frac{1}{2} (N^2 + 2N - 4N) = \frac{1}{2} N(N-2)$$

$$2 \left(\frac{N}{2} \right) (N-1) = N(N-1)$$

* Have twice as many values also no deg whif $\in Z_{\bar{a}}$ is

$$(2 \times 2) \cdot (2 \times N) = 2 \times 2N$$

$$\in Z_a = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \dots \\ \dots \end{pmatrix}$$

ex: $N=2, Z^{ab} = ZE^{ab}$

$$\{Q_{\alpha}^a, \bar{Q}_{\beta}^b\} = 2M \delta_{\alpha\beta} \delta^{ab}$$

$$\{Q_{\alpha}^a, Q_{\beta}^b\} = 2Z \epsilon_{\alpha\beta} E^{ab}$$

$$\{\bar{Q}_{\alpha}^a, \bar{Q}_{\beta}^b\} = 2Z \epsilon_{\alpha\beta} E^{ab}$$

On time $b_{\alpha}^{\pm} \equiv Q_{\alpha} \pm \epsilon_{\alpha\beta} Q_{\beta}$ $(b_{\alpha}^{\pm})^{\dagger} = Q_{\alpha} \pm \epsilon_{\alpha\beta} Q_{\beta}$

$$\begin{cases} \{b_{\alpha}^{+}, (b_{\beta}^{+})^{\dagger}\} = 4\delta_{\alpha\beta}(M+Z) \\ \{b_{\alpha}^{-}, (b_{\beta}^{-})^{\dagger}\} = 4\delta_{\alpha\beta}(M-Z) \end{cases}$$

$\therefore$ BPS Bound $M \geq Z$

Field Content & Lagrangians:

$A_{\mu}: 1$
 $\psi: 2/2$
 $\phi: 0$

* $\mathcal{G}$: Gauge Algebra associated w/ the Lie Group G .

$\forall 1 \leq N \leq 4$: Gauge Multiplets

(i) Gauge Mult: Transform in Adjoint of $\mathcal{G}$

(ii) Matter Multiplets:

$N=1$: Chiral Mult.

$N=2$: Hypermultiplet

A_{μ} : Gauge field

$\lambda_{\alpha}, \psi_{\alpha}$: Weyl

ϕ, H, X : Scalar fields

(i) $N=1$ Gauge Multiplet: $(A_{\mu}, \lambda_{\alpha})$

λ_{α} : Gaugino Majorana fermion

(ii) $N=2$ chiral Mult.: (ψ_{α}, ϕ)

ϕ : complex scalar

(iii) $N=2$ Gauge Multiplet: $(A_{\mu}, \lambda_{\alpha}^{\pm}, \phi)$

ϕ : complex gauge scalar

4. Complex Gauge scalar
 $\rightarrow$ Under $SU(2)_R$ A_m and ϕ are singlets
 λ_+ and λ_- transform as doublets.

(iv) $N=2$ Hyper $(\Psi_{\alpha\pm}, H_{\pm})$
 $H_+, H_- = \underline{2}$ of $SU(2)_R$

(v) $N=4$ Gauge Mult. (A_m, λ_a^a, X^i)

$a=1, \dots, 4$

λ_a^a : left Weyl fermions

$X^i, i=1, \dots, 6$ $\mathbb{R}$ -scalars

$\rightarrow SU(4)_R$: $A_m = \underline{1}$
 $\lambda_a^a = \underline{4}$

* Take the six scalars: X^i

$M_{ij} = -M_{ji}$ $\frac{n(n-1)}{2}$
 $n=4$: 6 unique elements.

$$[M_{ij}] = \begin{pmatrix} 0 & -X^1 & -X^2 & -X^3 \\ X^1 & 0 & -X^4 & -X^5 \\ X^2 & X^4 & 0 & -X^6 \\ X^3 & X^5 & X^6 & 0 \end{pmatrix}$$

$\{X^i\} \rightarrow [M_{ij}] = \underline{6}$ of $SU(4)_R$

* $N=1$ Gauge Multiplet (A_m, λ_a)

$$\mathcal{L} = -\frac{1}{2}g^2 \text{tr}(F_{\mu\nu}F^{\mu\nu}) + \frac{\theta_{\pm}}{8\pi^2} \text{tr}(F_{\mu\nu}\tilde{F}^{\mu\nu}) - \frac{i}{2} \text{tr}(\bar{\lambda} \not{D}_m \lambda)$$

g : Gauge coupling $|F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]$
 $\theta_{\pm}$: instanton angle

$$D_\mu \lambda = \partial_\mu \lambda + i[A_\mu, \lambda] \quad F = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}$$

$N=1$ susy transformations:

$$\delta_\epsilon A_\mu = i \bar{\epsilon} \bar{\sigma}_\mu \lambda - i \bar{\lambda} \bar{\sigma}_\mu \epsilon$$

$$\delta_\epsilon \lambda = \sigma^{\mu\nu} F_{\mu\nu} \lambda$$